Hybrid Machine Learning Models for Real-Time Detection of Financial Fraud in ERP Transaction Logs

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Abstract

Enterprise Resource Planning (ERP) systems serve as the core financial infrastructure of modern organizations by managing accounting, procurement, payroll, and transactional operations within a centralized digital environment. As ERP platforms increasingly handle high volumes of financial transactions, they have become attractive targets for fraudulent activities, including unauthorized payments, falsified invoices, and manipulation of accounting records. Traditional fraud detection mechanisms embedded in ERP systems are largely rule-based and reactive, limiting their effectiveness against complex and evolving fraud schemes. This research proposes a hybrid machine learning approach for real-time detection of financial fraud in ERP transaction logs. The proposed approach combines multiple machine learning techniques, including supervised and unsupervised learning models, to enhance detection accuracy and adaptability in dynamic enterprise environments. By integrating these models into a real-time transaction monitoring framework, the system continuously analyzes ERP transaction logs to identify suspicious patterns and generate immediate alerts. The hybrid model leverages the strengths of different algorithms to reduce false positives while improving fraud detection precision. This study also explores the architectural design required to support real-time analysis of ERP transaction streams and automated alert generation. Experimental evaluation using ERP-oriented financial datasets demonstrates that hybrid machine learning models outperform single-model approaches in detecting fraudulent transactions. The findings highlight the potential of hybrid AI techniques to improve financial security, strengthen internal controls, and support continuous auditing within ERP systems.

Keywords

Hybrid Machine Learning, ERP Systems, Financial Fraud Detection, Real-Time Monitoring, Transaction Logs, Artificial Intelligence.

Introduction

Background of ERP Transaction Processing

Enterprise Resource Planning systems are widely adopted by organizations to integrate and automate core business processes across finance, procurement, inventory, human resources, and operations. Within ERP platforms, transaction logs record every financial activity executed across different modules, creating a detailed and chronological record of enterprise financial behavior. These transaction logs include critical information such as transaction amounts, timestamps, user identifiers, account codes, approval workflows, and system actions. As organizations increasingly rely on ERP systems for financial decision-making and regulatory compliance, the integrity of transaction logs has become essential for maintaining trust and transparency. However, the high volume and complexity of ERP transaction logs present significant challenges for traditional monitoring and auditing approaches. Manual review of transaction logs is impractical in large organizations due to the sheer scale of data generated daily. Additionally, static control rules embedded in ERP systems often fail to capture subtle or emerging fraud patterns that exploit system loopholes. Fraudulent activities may be concealed within normal transaction flows, making detection difficult without advanced analytical tools. As ERP environments continue to evolve and transaction volumes increase, there is a growing need for intelligent systems capable of analyzing transaction logs in real time and identifying suspicious behaviors automatically. Machine learning techniques offer promising solutions by enabling automated pattern recognition and adaptive learning from historical transaction data.

Problem Statement

Despite the critical role of ERP systems in managing enterprise financial operations, financial fraud remains a persistent challenge that undermines organizational integrity and financial stability. ERP transaction logs contain valuable information that could be used to detect fraudulent behavior; however, existing fraud detection mechanisms are often limited in scope and effectiveness. Traditional rule-based systems rely on predefined thresholds and static business rules that require frequent manual updates to remain relevant. These systems are particularly vulnerable to sophisticated fraud schemes that evolve over time or deliberately bypass known control rules. Furthermore, many existing fraud detection solutions operate in batch-processing modes, analyzing transaction logs only after financial periods have closed. This delayed detection approach allows fraudulent activities to persist undetected, increasing potential financial losses and reputational damage. Single-model machine learning approaches, while more flexible than rule-based systems, may also struggle to capture the full complexity of ERP transaction data due to the diversity of transaction types and behavioral patterns. Some models may excel at detecting known fraud patterns but perform poorly when encountering new or rare anomalies. Additionally, high false positive rates can overwhelm auditors with excessive alerts, reducing system usability. These challenges highlight the need for a more robust and adaptive fraud detection approach. Therefore, there is a critical need for hybrid machine learning models that combine multiple analytical techniques to improve real-time fraud detection accuracy and resilience within ERP transaction logs.

Research Objectives

The primary objective of this research is to develop and evaluate a hybrid machine learning framework for real-time detection of financial fraud in ERP transaction logs. The study aims to design a system that integrates multiple machine learning models to leverage their complementary strengths in identifying fraudulent transaction patterns. One key objective is to combine supervised learning techniques, which are effective at detecting known fraud scenarios, with unsupervised learning methods capable of identifying previously unseen anomalies within transaction data. Another objective is to design a real-time monitoring architecture that enables continuous analysis of ERP transaction logs as financial events occur. The research also seeks to reduce false positive rates commonly associated with fraud detection systems by improving model robustness and decision accuracy through hybridization. Additionally, the study aims to evaluate the performance of hybrid models using standard fraud detection metrics such as precision, recall, and detection latency. The research further aims to contribute to the understanding of how hybrid machine learning approaches can enhance financial monitoring and auditing processes in enterprise environments. By achieving these objectives, the study intends to provide organizations with a scalable and intelligent solution for strengthening financial security, improving fraud detection efficiency, and supporting continuous auditing practices within ERP systems.

Research Questions

This study is guided by several research questions that explore the effectiveness of hybrid machine learning models in detecting financial fraud within ERP transaction logs. The first research question examines how hybrid machine learning approaches can improve fraud detection accuracy compared to single-model techniques when applied to ERP transaction data. Understanding the comparative performance of hybrid models is essential for validating their practical value in enterprise environments. The second research question focuses on identifying which combinations of supervised and unsupervised learning algorithms are most effective for real-time fraud detection in ERP systems. This question seeks to evaluate how different model pairings complement each other in identifying both known and unknown fraud patterns. The third research question investigates how real-time transaction monitoring can be implemented within ERP systems without disrupting core business operations. Another research question explores how hybrid models impact false positive rates and auditor workload in financial monitoring systems. Finally, the study seeks to understand the implications of deploying hybrid fraud detection models on organizational financial governance and internal control effectiveness. These research questions provide a structured framework for evaluating the proposed approach and assessing its contribution to ERP-based financial fraud detection research.

Significance of the Study

The significance of this study lies in its contribution to improving financial fraud detection capabilities within ERP systems through the application of hybrid machine learning techniques. Financial fraud poses serious risks to organizations, including financial losses, regulatory penalties, and reputational damage. By proposing a real-time fraud detection framework based on hybrid models, this research offers a proactive approach to identifying suspicious transactions before significant harm occurs. The study is particularly valuable for large enterprises that process high volumes of financial transactions and require automated monitoring solutions capable of scaling with organizational growth. From a practical perspective, the proposed approach can help reduce auditor workload by minimizing false alerts and prioritizing high-risk transactions for investigation. The framework also supports continuous auditing practices, enabling organizations to strengthen internal controls and improve compliance with financial regulations. From a theoretical standpoint, the study contributes to the growing body of research on machine learning applications in enterprise information systems by demonstrating how hybrid approaches can enhance fraud detection performance. Additionally, the findings provide insights for ERP vendors and system developers seeking to incorporate intelligent fraud detection capabilities into future ERP platforms.

Structure of the Paper

This paper is organized into several sections to present the research in a clear and systematic manner. The first section introduces the research topic by discussing the background of ERP transaction processing, the problem of financial fraud, and the objectives and significance of the study. The second section presents a comprehensive literature review covering ERP transaction logs, traditional fraud detection techniques, machine learning applications in fraud detection, and hybrid modeling approaches. This section also identifies limitations in existing studies and highlights the research gap addressed by this work. The third section describes the research methodology, including data sources, preprocessing techniques, hybrid model design, system architecture, and evaluation metrics. The fourth section presents the proposed hybrid machine learning framework for real-time fraud detection and explains its components and operational workflow. The fifth section reports experimental results and analyzes the performance of the hybrid models. The sixth section discusses the findings and their implications for ERP-based fraud detection. The final sections conclude the paper by summarizing key contributions, outlining limitations, and suggesting directions for future research.

Literature Review

ERP Transaction Logs and Financial Fraud

Enterprise Resource Planning transaction logs serve as comprehensive digital records that capture every operational and financial activity occurring within an organization's integrated information system. These logs include detailed information such as transaction timestamps, user identities, transaction values, system actions, account codes, and approval processes. Because ERP systems manage critical financial operations including procurement payments, payroll processing, invoice management, and asset accounting, the transaction logs generated by these systems represent valuable data sources for monitoring financial behavior and identifying irregular activities. Financial fraud within ERP environments can occur in multiple forms, including unauthorized payments, vendor manipulation, duplicate invoices, ghost employee payroll entries, and internal financial misappropriation. Such fraudulent activities may be concealed within normal transaction flows, making them difficult to detect using conventional auditing methods. As the volume of ERP transactions continues to grow due to digital transformation and global business expansion, manual analysis of transaction logs becomes increasingly impractical and inefficient. Traditional monitoring approaches typically rely on periodic reviews conducted by auditors, which may allow fraudulent transactions to remain undetected for extended periods. Additionally, rule-based monitoring systems embedded in ERP platforms are often limited to predefined thresholds that fail to capture complex fraud schemes. As a result, researchers have increasingly emphasized the need for advanced analytical techniques capable of analyzing ERP transaction logs in real time. Machine learning technologies offer significant potential for improving fraud detection by automatically identifying abnormal patterns and behaviors within large transactional datasets. These intelligent systems can continuously learn from historical transaction data and detect deviations that may indicate fraudulent activities, thereby strengthening financial governance and internal control mechanisms within enterprise systems.

Traditional Fraud Detection Techniques

Traditional fraud detection techniques used in financial systems primarily rely on rule-based monitoring mechanisms and statistical analysis methods designed to identify irregular transaction patterns. Rule-based systems operate by defining specific business rules, thresholds, or control parameters that trigger alerts when transactions violate established policies. For example, ERP systems may generate warnings when transaction values exceed predefined limits, when payments are processed outside normal working hours, or when unusual vendor accounts are involved in financial activities. While such rule-based approaches are relatively easy to implement and interpret, they possess significant limitations in detecting sophisticated fraud schemes. Fraudulent actors often adapt their behaviors to remain within acceptable rule boundaries, thereby avoiding detection. Additionally, rule-based systems require continuous updates and maintenance to remain effective as organizational processes and fraud tactics evolve over time. Statistical fraud detection techniques represent another traditional approach used to analyze deviations from expected financial behavior. These methods apply statistical models such as regression analysis, probability distributions, and outlier detection to identify transactions that differ significantly from historical norms. Although statistical models can detect certain anomalies, they often struggle to handle the high dimensionality and complexity of ERP transaction data. Furthermore, these models may not adapt easily to rapidly changing financial patterns within dynamic enterprise environments. As organizations increasingly process millions of financial transactions through ERP systems, the limitations of traditional fraud detection approaches become more apparent. These challenges have motivated the exploration of advanced analytical methods such as machine learning, which can automatically learn complex transaction patterns and improve fraud detection capabilities through adaptive data-driven approaches.

Machine Learning in Financial Fraud Detection

Machine learning has emerged as one of the most promising technological approaches for detecting financial fraud in modern digital systems due to its ability to analyze complex datasets and identify hidden patterns within large volumes of transactional information. Unlike traditional rule-based systems that depend on predefined conditions, machine learning algorithms learn patterns directly from historical data and use these insights to predict future behaviors or detect anomalies. In financial fraud detection, machine learning models are commonly categorized into supervised and unsupervised learning approaches depending on the availability of labeled data. Supervised learning models are trained using datasets that contain examples of both legitimate and fraudulent transactions. Algorithms such as decision trees, random forests, logistic regression, and support vector machines can learn the distinguishing characteristics of fraudulent activities and classify new transactions accordingly. These models are particularly effective in detecting known fraud scenarios where historical examples are available. In contrast, unsupervised learning techniques do not rely on labeled datasets but instead identify anomalies by detecting deviations from normal transaction patterns. Methods such as clustering algorithms, isolation forests, and autoencoders are widely used for anomaly detection tasks in financial systems. Machine learning approaches offer several advantages over traditional techniques, including improved detection accuracy, scalability for large datasets, and the ability to adapt to evolving fraud patterns. However, individual machine learning models may still face limitations when applied to complex ERP environments, as they may struggle to capture the full diversity of transaction behaviors present in integrated enterprise systems.

Hybrid Machine Learning Approaches

Hybrid machine learning approaches have gained increasing attention in fraud detection research because they combine the strengths of multiple algorithms to improve overall detection performance and reliability. In many cases, a single machine learning model may not be sufficient to capture all types of fraudulent behavior present within complex financial systems. Supervised models may perform well when detecting known fraud patterns but struggle when encountering new or previously unseen anomalies. Conversely, unsupervised models are capable of identifying novel anomalies but may produce higher false positive rates because they lack labeled training data. Hybrid approaches address these limitations by integrating different types of algorithms into a unified analytical framework. For example, supervised classification models may be combined with unsupervised anomaly detection techniques to provide both predictive accuracy and anomaly discovery capabilities. Ensemble learning methods, which aggregate the predictions of multiple models, are commonly used to improve robustness and reduce detection errors. In the context of ERP transaction log analysis, hybrid machine learning models can analyze transaction behaviors from multiple perspectives, enabling more comprehensive fraud detection. These systems may utilize layered architectures in which one model filters suspicious transactions while another model performs deeper anomaly analysis. By combining multiple analytical techniques, hybrid approaches can achieve higher detection accuracy, lower false positive rates, and greater adaptability to evolving fraud strategies. Consequently, hybrid machine learning models are increasingly recognized as effective solutions for enhancing financial fraud detection in enterprise systems.

Limitations of Existing Studies

Although significant progress has been made in the development of machine learning-based fraud detection systems, several limitations remain in existing research, particularly when applied to ERP transaction log environments. Many studies focus on financial datasets obtained from banking systems or credit card transactions rather than integrated enterprise resource planning platforms. As a result, the proposed models may not fully capture the complexity and diversity of transaction behaviors present in ERP systems. ERP environments generate highly heterogeneous data that includes interactions across multiple modules such as procurement, payroll, asset management, and financial accounting. This multi-module structure introduces additional challenges for fraud detection models that were originally designed for simpler financial datasets. Another limitation of existing studies is the lack of real-time processing capabilities. Many fraud detection models are evaluated using static datasets processed offline, which does not accurately reflect the operational requirements of enterprise environments where immediate detection is critical. Additionally, some machine learning models require extensive computational resources and may not scale efficiently when applied to large volumes of ERP transaction data. High false positive rates also remain a common challenge, as excessive alerts can overwhelm auditors and reduce the practical usefulness of fraud detection systems.

Furthermore, limited attention has been given to system integration issues, including how machine learning models can be embedded within ERP infrastructures without disrupting normal business operations. These limitations highlight the need for more robust, scalable, and integrated fraud detection frameworks.

Research Gap

Based on the analysis of existing literature, a clear research gap can be identified in the development of hybrid machine learning frameworks specifically designed for real-time fraud detection within ERP transaction logs. While many studies have explored the use of machine learning algorithms for financial fraud detection, relatively few have addressed the unique challenges associated with ERP environments, including the complexity of transaction data, the integration of multiple operational modules, and the need for continuous monitoring. Furthermore, most existing research focuses on single-model approaches rather than hybrid frameworks that combine the advantages of multiple machine learning techniques. There is also limited research examining how hybrid models can be implemented within real-time monitoring architectures capable of analyzing ERP transaction streams as they occur. Another gap exists in the integration of fraud detection models with automated alerting and auditing mechanisms that support continuous financial oversight. Many studies evaluate algorithmic performance using isolated datasets without considering how the models would function within operational enterprise systems. This research seeks to address these gaps by proposing a hybrid machine learning framework specifically tailored for ERP transaction log analysis and real-time fraud detection. The proposed approach integrates supervised and unsupervised learning models within a scalable monitoring architecture designed to process high volumes of transaction data while maintaining high detection accuracy and operational efficiency.

Methodology

Research Design

This study adopts a **design science and experimental research approach** to develop a hybrid machine learning framework for real-time financial fraud detection within ERP systems. The research design is focused on both the theoretical and practical aspects of creating a robust fraud detection system that can operate under realistic enterprise conditions. The design science approach is particularly suitable because it emphasizes the iterative creation and evaluation of artifacts—in this case, the hybrid machine learning framework—while addressing a specific organizational problem: fraudulent activities hidden within ERP transaction logs. The study follows a systematic methodology that involves identifying functional requirements, selecting appropriate machine learning models, designing a real-time monitoring architecture, and integrating alerting and audit mechanisms. An experimental component is included to assess the performance of the hybrid models using ERP-like datasets, measuring detection accuracy, false positive rates, and processing efficiency. The research emphasizes scalability, adaptability, and system integration to ensure that the framework can operate efficiently within large organizations handling millions of daily transactions. Additionally, the iterative design allows continuous refinement of the hybrid models to address evolving financial fraud patterns, ensuring that the framework remains relevant over time. By combining design science principles with empirical evaluation, this research ensures that the proposed framework is not only theoretically sound but also practically deployable, offering a comprehensive solution for improving financial monitoring and fraud prevention in enterprise environments.

Data Sources

The data sources utilized in this study consist primarily of ERP transaction log datasets that simulate real-world financial operations across multiple enterprise modules, including accounting, procurement, payroll, and asset management. Each transaction record contains critical attributes such as transaction ID, timestamp, user ID, transaction type, account code, transaction amount, and approval status. To create realistic scenarios, both **legitimate and fraudulent transactions** are included, reflecting common fraud schemes such as duplicate payments, ghost employees, unauthorized transfers, and vendor collusion. In cases where access to confidential enterprise ERP data is restricted, simulated datasets are generated using statistical methods and historical transaction distributions to replicate real-world ERP behavior. The dataset is divided into training, validation, and testing subsets to ensure proper machine learning model development and evaluation. The training subset is used to build predictive and anomaly detection models, while the validation subset fine-tunes model parameters. The testing subset evaluates model performance on unseen data to simulate deployment scenarios. Using multi-module ERP datasets allows the study to capture the complex, interdependent patterns inherent in enterprise financial operations, which is essential for effective fraud detection. Additionally, including both known fraud instances and anomalous patterns supports the evaluation of hybrid machine learning models in detecting both **prevalent and novel fraud types**, providing a comprehensive assessment of their real-time applicability within ERP systems.

Data Preprocessing

Data preprocessing is a critical step in preparing ERP transaction logs for hybrid machine learning analysis, ensuring that models can accurately detect fraudulent activity while handling high-dimensional and complex datasets. Preprocessing begins with **data cleaning**, which involves removing duplicate entries, correcting inconsistent formats, and imputing missing values to maintain data integrity. Categorical attributes such as transaction type, account code, and user role are encoded into numerical representations suitable for machine learning algorithms, while numerical features such as transaction amounts are normalized to eliminate scale biases. **Feature extraction** techniques are applied to derive additional indicators of potential fraud, such as transaction frequency per user, unusual transaction amounts relative to historical patterns, and atypical time-of-day activity. **Feature selection** methods are employed to identify the most informative variables for anomaly detection, reducing dimensionality and improving computational efficiency. Temporal features are particularly important for ERP logs, as time-based anomalies, including transactions outside normal business hours or repeated suspicious patterns over short periods, can indicate fraudulent behavior. The preprocessed dataset is structured to support **real-time streaming analysis**, allowing the hybrid machine learning framework to process incoming transactions efficiently and continuously. Proper preprocessing enhances model performance by improving the detection of subtle anomalies and reducing the likelihood of false positives, which is critical for operational deployment in enterprise environments where accuracy and reliability are paramount.

Hybrid Machine Learning Models

The hybrid machine learning component of the framework integrates **supervised and unsupervised algorithms** to detect both known and novel fraudulent patterns within ERP transaction logs. Supervised models, including decision trees, random forests, and gradient boosting classifiers, are trained using labeled datasets containing both legitimate and fraudulent transactions. These models excel at detecting previously observed fraud scenarios by learning distinguishing characteristics and classification rules. Unsupervised models, such as isolation forests, autoencoders, and clustering algorithms, complement supervised techniques by identifying anomalous transactions that deviate from established behavioral patterns, even in the absence of explicit labels. The hybrid approach combines outputs from these models using **ensemble methods** or weighted scoring to produce a unified fraud risk score for each transaction. Transactions exceeding predefined risk thresholds are flagged for further review. The combination of supervised and unsupervised methods mitigates the weaknesses of individual models: supervised models handle known fraud efficiently but may miss new anomalies, while unsupervised models detect novel fraud but are prone to higher false positive rates. By integrating these approaches, the hybrid system achieves higher detection accuracy, reduced false positives, and adaptability to evolving fraud tactics. The hybrid design also supports scalability for high-volume ERP environments, enabling continuous monitoring of millions of daily transactions while maintaining real-time responsiveness and operational efficiency.

System Architecture Design

The proposed framework employs a **modular system architecture** that supports real-time ERP transaction monitoring and hybrid fraud detection. The architecture comprises several key layers: the **data ingestion layer**, responsible for streaming ERP transaction logs in real time; the **data preprocessing layer**, which performs cleaning, normalization, and feature engineering; the **hybrid analysis layer**, where supervised and unsupervised machine learning models operate in parallel and their outputs are combined; the **alert management layer**, which generates notifications for flagged suspicious transactions; and the **audit and reporting layer**, which maintains an immutable record of detected anomalies, risk scores, and system actions. The modular design allows for **scalable deployment** within large enterprises, enabling seamless integration with existing ERP infrastructures without disrupting core operations. Each layer communicates through secure APIs or message queues to ensure real-time performance, data integrity, and minimal latency. The architecture also supports dynamic model updates, allowing the hybrid models to retrain periodically as new transaction patterns and fraud schemes emerge. Visualization dashboards provide auditors and financial managers with insights into anomalies, fraud trends, and system performance metrics. This architecture ensures that the framework can process high volumes of transaction data efficiently while providing accurate, timely, and explainable fraud detection results.

Evaluation Metrics

To assess the performance of the hybrid machine learning framework, multiple evaluation metrics are employed, focusing on both predictive accuracy and operational effectiveness. **Accuracy** measures the proportion of correctly classified transactions among all transactions analyzed, providing a general indication of model performance. **Precision** quantifies the fraction of true fraudulent transactions among those flagged as suspicious, which is critical for minimizing false alerts that could overwhelm auditors. **Recall** evaluates the model's ability to identify all actual fraudulent transactions, ensuring that critical cases are not missed. The **F1-score** combines precision and recall to provide a balanced metric reflecting both detection quality and reliability. Additionally, the **false positive rate** is analyzed to assess the system's operational efficiency, as high false positives can undermine trust in automated monitoring. **Detection latency** is measured to evaluate real-time responsiveness, representing the time taken to process a transaction and generate an alert. By examining these metrics, the framework's ability to accurately detect fraud, maintain low false alarm rates, and operate in real-time ERP environments is rigorously evaluated. This comprehensive assessment ensures that the proposed hybrid system meets both technical and operational requirements for effective enterprise fraud detection.

Proposed Hybrid Fraud Detection Framework

Framework Overview

The proposed hybrid fraud detection framework is designed to provide **real-time monitoring and intelligent analysis** of ERP transaction logs, integrating multiple machine learning models to detect both known and emerging fraudulent behaviors. The framework operates as a multi-layered system, enabling seamless data ingestion, preprocessing, hybrid analysis, alert generation, and audit logging within enterprise environments. At its core, the system leverages the strengths of supervised models to classify previously observed fraud patterns and unsupervised models to identify novel anomalies that deviate from normal transaction behavior. Transaction logs are captured continuously from ERP modules, including accounting, procurement, payroll, and asset management, ensuring a holistic view of enterprise financial operations. Preprocessing transforms raw transactional data into structured feature representations suitable for analysis, including temporal patterns, transaction frequency, and statistical deviations. The hybrid analysis layer applies ensemble techniques to integrate the outputs of multiple models, producing a unified fraud risk score for each transaction. Transactions exceeding predefined thresholds trigger automated alerts and notifications to auditors, who can investigate high-risk activities promptly. Additionally, an audit trail and reporting module records all detected anomalies, model decisions, and system actions in a tamper-resistant format, ensuring transparency and compliance with regulatory standards. This framework is modular and scalable, allowing organizations to deploy it without disrupting existing ERP operations. By combining **real-time monitoring, hybrid machine learning, and automated reporting**, the framework addresses the limitations of conventional fraud detection systems, providing a comprehensive and adaptive solution capable of responding to evolving fraud strategies while maintaining operational efficiency.

Real-Time Transaction Log Processing

Real-time transaction log processing is a critical component of the proposed framework, ensuring that financial activities within ERP systems are continuously monitored for potential fraudulent behaviors. ERP platforms generate a high volume of transactions across multiple modules, and any delay in detection can result in financial losses or compliance violations. To address this, the framework employs streaming data pipelines that capture transactions as they occur, using secure APIs or message queues to transfer log data to the preprocessing and hybrid analysis layers. Each transaction is parsed and structured to extract essential features such as transaction amount, user identity, account codes, transaction type, timestamp, and approval hierarchy. Temporal and behavioral patterns, including unusual activity outside business hours, repeated high-value transactions, or deviations from historical patterns, are computed to enhance anomaly detection. Real-time processing also incorporates data buffering and batch handling to ensure that the system remains responsive under high transaction loads, preventing bottlenecks that could delay alerts. By continuously feeding transactions into the hybrid machine learning models, the system identifies suspicious activities immediately, rather than relying on periodic batch analysis. This approach improves the responsiveness of fraud detection, allowing organizations to take timely action against potentially fraudulent transactions. The integration of real-time processing with the hybrid detection engine ensures that the framework can operate efficiently in dynamic enterprise environments, handling millions of transactions daily while maintaining high detection accuracy and operational reliability.

Hybrid Model Integration Layer

The hybrid model integration layer is the analytical core of the proposed framework, combining the outputs of multiple machine learning algorithms to achieve accurate and robust fraud detection. This layer integrates both **supervised models**, which classify transactions based on historical labeled data, and **unsupervised models**, which identify anomalies that deviate from normal patterns without requiring explicit labels. Ensemble techniques such as weighted averaging, voting mechanisms, or stacking are applied to merge the outputs of individual models into a unified fraud risk score for each transaction. By leveraging complementary strengths, the hybrid layer mitigates the weaknesses of individual algorithms: supervised models are effective for known fraud scenarios but may miss novel patterns, while unsupervised models can detect previously unseen anomalies but may generate higher false positive rates. Feature importance analysis is performed to understand which transaction attributes most strongly influence model decisions, improving explainability for auditors and regulatory compliance purposes. Periodic retraining is implemented to ensure that models adapt to evolving transaction patterns and emerging fraud tactics. The hybrid integration layer also incorporates thresholds and confidence scoring to prioritize transactions for further investigation, ensuring that high-risk activities are immediately flagged while maintaining low false positive rates. This approach ensures that the framework is both **accurate and operationally efficient**, capable of handling the volume, velocity, and complexity of ERP transaction logs while providing actionable insights for enterprise financial governance.

Alert and Response Mechanism

The alert and response mechanism in the proposed framework ensures timely notification and investigation of suspicious transactions identified by the hybrid machine learning models. When a transaction exceeds the predefined fraud risk threshold, the system automatically generates an alert containing critical details, including transaction identifiers, timestamp, risk score, and relevant contextual information such as the affected ERP module, user role, and approval hierarchy. These alerts are transmitted through secure channels to auditors, financial managers, or compliance officers, enabling rapid assessment and intervention. The framework supports **prioritization of alerts** based on risk severity, ensuring that high-risk transactions are investigated immediately, while lower-risk anomalies are reviewed in aggregated reports. Automated response workflows can also be implemented, including temporary transaction holds, notifications to relevant departments, or escalation protocols for repeated or high-value fraudulent activities. Integration with dashboard visualization tools provides auditors with comprehensive views of real-time fraud trends, transaction distributions, and historical anomaly patterns, improving situational awareness and decision-making efficiency. This continuous and responsive alert system reduces the latency between fraud occurrence and detection, minimizing potential financial damage and enhancing internal controls. By combining automated notifications with human oversight, the framework ensures a **balanced approach** that leverages the efficiency of AI while maintaining accountability and compliance standards within enterprise financial operations.

Audit Trail and Reporting Module

The audit trail and reporting module ensures **transparency, traceability, and regulatory compliance** by recording all actions performed within the hybrid fraud detection framework. Every transaction flagged as suspicious, along with its corresponding risk score, model outputs, and system-generated alerts, is logged in a secure, immutable audit trail. This comprehensive record includes timestamps, user interactions, and investigation outcomes, providing a detailed chronological sequence that auditors can review during investigations. The reporting module generates periodic and ad hoc analytical reports summarizing transaction anomalies, fraud trends, detection performance, and system efficiency. Visualization tools, including charts, dashboards, and heatmaps, allow financial managers to interpret complex data intuitively, identify systemic fraud patterns, and monitor risk exposure across multiple ERP modules. Audit trail data also supports internal and external regulatory audits, providing proof of continuous monitoring and fraud mitigation practices. The module is designed to comply with data protection and financial governance standards, ensuring that sensitive financial information is securely stored and accessible only to authorized personnel. By automating recordkeeping and reporting, this module reduces the administrative burden on auditors, enhances operational transparency, and strengthens organizational confidence in the integrity of ERP financial monitoring systems.

Experimental Results and Analysis

Dataset Description

The experimental evaluation of the proposed hybrid fraud detection framework utilizes ERP-oriented financial transaction datasets designed to simulate realistic enterprise financial operations. These datasets include transactional records generated across multiple ERP modules, such as accounting, procurement, payroll, vendor management, and asset management. Each record contains detailed attributes including transaction ID, timestamp, transaction type, amount, associated account, user role, and approval workflow, providing a comprehensive representation of typical ERP activity. To enable the assessment of fraud detection accuracy, the dataset incorporates both legitimate transactions and synthetic fraudulent transactions reflecting common fraud scenarios, including duplicate payments, ghost employees, unauthorized transfers, and vendor collusion. Fraudulent transactions are embedded in realistic proportions relative to overall transaction volume, simulating conditions that enterprises encounter in real operational environments. The dataset is partitioned into training, validation, and testing subsets to support hybrid machine learning development and performance evaluation. The training set is used to establish supervised and unsupervised models, while the validation set fine-tunes model parameters. The testing set evaluates model performance on unseen data, reflecting real deployment conditions. By including diverse transaction types, multiple ERP modules, and realistic fraud patterns, the dataset enables comprehensive evaluation of the hybrid framework's ability to detect both known and previously unseen fraudulent activities. This structured and high-fidelity dataset ensures that experimental results are representative of actual enterprise environments, supporting robust conclusions regarding detection accuracy, real-time responsiveness, and operational scalability.

Implementation Environment

The hybrid fraud detection framework is implemented using a combination of **Python-based machine learning libraries** and data processing tools suitable for high-volume ERP transaction analysis. Supervised models, including decision trees, random forests, and gradient boosting classifiers, are implemented using Scikit-learn and XGBoost, while unsupervised models, including isolation forests and autoencoders, are implemented with Scikit-learn and TensorFlow. Data preprocessing and feature engineering are performed using Pandas and NumPy, ensuring efficient handling of large-scale transaction datasets. The system is deployed in a modular architecture with separate layers for data ingestion, preprocessing, hybrid analysis, alerting, and audit logging, facilitating scalability and maintainability. Real-time transaction streaming is simulated using Python message queues and batch pipelines, enabling evaluation of latency and responsiveness. Visualization and reporting are implemented using Matplotlib, Seaborn, and dashboard frameworks to monitor anomalies and performance metrics. The implementation environment includes hardware with high-memory CPUs and GPUs for efficient training of complex models, allowing evaluation of computational efficiency alongside detection accuracy. Model training incorporates cross-validation and hyperparameter tuning to optimize performance, while testing evaluates both classification metrics and real-time operational feasibility. By leveraging widely adopted machine learning and data processing technologies, the framework demonstrates practical applicability for enterprise deployment and provides a realistic platform for assessing hybrid fraud detection performance in ERP transaction log environments.

Performance Evaluation

The hybrid fraud detection framework is evaluated using multiple performance metrics to assess both predictive accuracy and operational effectiveness. Key metrics include **accuracy**, representing the proportion of correctly classified transactions; **precision**, quantifying the proportion of true fraudulent transactions among those flagged as suspicious; **recall**, measuring the ability of the system to detect all actual fraudulent transactions; and **F1-score**, balancing precision and recall to provide a holistic assessment of detection performance. The **false positive rate** is also analyzed, as excessive false alerts can overwhelm auditors and reduce operational efficiency. In addition to these metrics, **detection latency** is evaluated to measure the framework's ability to process transactions and generate alerts in real time. Experimental results indicate that the hybrid model significantly outperforms individual supervised or unsupervised models in both detection accuracy and false positive reduction. Ensemble methods effectively combine outputs from multiple models, enhancing robustness against diverse fraud patterns. The framework demonstrates consistent performance across multiple ERP modules, including payroll, procurement, and accounting, while maintaining real-time responsiveness suitable for operational deployment. Comparative analyses show that the hybrid system achieves higher recall and precision than single-model approaches, highlighting its ability to detect both known and novel fraud patterns. The results validate the effectiveness of the proposed framework in accurately identifying suspicious transactions, supporting timely alerts, and providing actionable insights for auditors and financial managers in complex ERP environments.

Discussion

The experimental results demonstrate that the proposed hybrid machine learning framework significantly improves real-time financial fraud detection in ERP transaction logs compared to conventional single-model or rule-based approaches. The integration of supervised and unsupervised algorithms allows the system to detect both previously observed fraud patterns and novel anomalies that would otherwise evade traditional detection mechanisms. High accuracy, precision, and recall metrics indicate that the hybrid approach effectively identifies suspicious transactions while minimizing false positives, reducing the workload on auditors and improving operational efficiency. The modular system architecture supports real-time data ingestion and analysis across multiple ERP modules, ensuring that fraudulent activity can be detected as it occurs, rather than after delays associated with batch processing. The results also highlight the importance of comprehensive feature engineering, including temporal, behavioral, and transactional indicators, in enhancing model performance and interpretability. By producing a unified risk score through ensemble integration, the framework provides actionable insights for auditors and financial managers, facilitating rapid decision-making and timely intervention. Furthermore, the audit trail and reporting module ensures transparency and compliance by maintaining detailed records of all detected anomalies and system actions. These findings suggest that hybrid machine learning techniques are particularly effective in complex, high-volume ERP environments, addressing the limitations of both rule-based and single-model machine learning approaches. Overall, the study underscores the potential of combining multiple algorithms within a real-time monitoring architecture to enhance enterprise financial security, strengthen internal controls, and support continuous auditing practices.

Implications of the Study

This research carries significant implications for both theory and practice in enterprise financial monitoring. From a theoretical perspective, the study contributes to the growing body of knowledge on hybrid machine learning applications in fraud detection, demonstrating how combining supervised and unsupervised models enhances detection accuracy and adaptability. The findings also provide insights into model integration, ensemble scoring, and feature selection strategies that are effective in real-time ERP environments. Practically, the proposed framework offers organizations a scalable and operationally efficient solution for continuous fraud monitoring, reducing reliance on manual auditing processes and enhancing risk management. By integrating real-time transaction log analysis, automated alerts, and transparent audit trails, organizations can detect suspicious activity promptly, minimizing financial losses and regulatory exposure. ERP vendors may leverage these findings to incorporate hybrid fraud detection capabilities into future system releases, providing built-in intelligent monitoring features. Additionally, compliance officers and auditors can benefit from improved decision support tools that prioritize high-risk transactions and reduce false alarms. The framework's modular architecture enables adaptation to different organizational contexts, ensuring that both large enterprises with high transaction volumes and smaller organizations can implement intelligent fraud detection effectively. Overall, the study emphasizes the potential of hybrid machine learning approaches to strengthen financial governance, improve operational efficiency, and enhance the overall security of enterprise financial systems.

Limitations and Future Research

Despite the promising results, several limitations of this study should be acknowledged. First, the experimental evaluation primarily relies on simulated ERP transaction datasets, which, although designed to be realistic, may not fully capture the diversity and complexity of real-world enterprise financial operations. Access to large-scale, real-world ERP transaction logs would allow more comprehensive validation of model performance. Second, while the hybrid approach reduces false positives compared to single models, some false alerts remain, potentially requiring manual investigation and increasing auditor workload in high-volume environments. Third, computational complexity may pose challenges in extremely large-scale ERP deployments, necessitating optimized model architectures or distributed processing frameworks. Additionally, the interpretability of hybrid models, particularly ensembles combining multiple algorithms, may be limited, making it difficult for auditors to understand specific model decisions. Future research could address these limitations by incorporating **explainable AI (XAI)** techniques to enhance interpretability, evaluating the framework on real-world ERP data from diverse organizations, and exploring distributed or cloud-based deployments for large enterprises. Further studies could also investigate the integration of blockchain technologies to enhance audit trail security and data immutability. Finally, research could examine adaptive model retraining techniques to maintain performance in dynamically evolving transaction environments, ensuring sustained effectiveness of hybrid fraud detection systems in operational ERP deployments.

Conclusion

This study presents a **hybrid machine learning framework for real-time financial fraud detection** in ERP transaction logs, addressing the limitations of traditional rule-based and single-model detection systems. By integrating supervised and unsupervised algorithms within a modular architecture, the framework effectively identifies both known and novel fraudulent activities while maintaining operational efficiency and minimizing false positives. Real-time transaction log processing, automated alerting, and a transparent audit trail provide organizations with actionable insights, timely intervention capabilities, and compliance support. Experimental evaluation using simulated ERP datasets demonstrates that hybrid models achieve superior detection accuracy, recall, and precision compared to individual models, validating the framework's effectiveness in complex enterprise environments. The study highlights the importance of comprehensive feature engineering, ensemble model integration, and continuous monitoring in enhancing financial governance and internal control mechanisms. While limitations exist, including reliance on simulated data and computational complexity, the framework provides a scalable, adaptable, and practical solution for enhancing ERP system security. Future research can expand on this work by incorporating real-world ERP data, explainable AI techniques, and distributed deployment strategies. Overall, this research demonstrates that hybrid machine learning approaches represent a powerful and feasible solution for strengthening financial oversight, mitigating fraud risk, and supporting continuous auditing practices in modern enterprise resource planning systems, contributing both to the academic literature and to practical enterprise financial governance.

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